1 Scope

Which entries of the formalizing registers concern homomorphic encryption, HE-OFT or the realization of releases by threshold CKKS, and where does each stand? The registers moved to FORMAL_SECURITY on 2026-11-11. Every entry of their problem, result and fact registers at HE-ML commit bba2cf1 was read, and an entry was kept when it named one of the three.

2 Result

HE-ML carries 5 open problems, 7 results and 6 facts on encryption, and none settles that threshold CKKS realizes an ideal release, which keeps reduction to CKKS heuristic. Against a semi-honest coalition, reduction to CKKS rests on thm-dataflow-composition, proven given the realization error, and on HE-OFT's simulation of a server with fewer than t clients. The simulation assumes a smudging that HE-OFT's measured configuration does not meet, and HHKLS26 is recorded, unread, as refuting per-party smudging, which HE-OFT and HLSS25 both use. Against a deviating coalition reduction to CKKS lacks support, because active security was found only for BFV, BGV and FHEW decryption and HE-OFT admits no deviating server. Steering lies outside reduction to CKKS, because a client of HE-OFT that inflates its encrypted per-class counts moves a row of the shared head at will (prop-he-oft-count-steering, proven). In the scenario space, reduction to CKKS thus covers semi-honest servers and clients heuristically, and the cells of deviating servers and of steering need another route or are gaps.

Does not show: Every HE-OFT result is as KSK26 states it, unchecked here. The two facts on per-party smudging rest on a sweep coder's extraction, and the search on active security read search summaries only. The entries are a snapshot of 2026-11-11.

3 How it was produced

Sources	the problem, result and fact registers in `FORMAL_SECURITY/HE-ML-migrated/sink/formalizing/`, copied from HE-ML commit bba2cf1
Range	entries written from 2026-10-16 to 2026-10-25
Screened	651 entries read, 18 kept
Artifacts	one CSV per table and a facts CSV in `lib/reports/src/data/`, under this report's stem
Reproduce	`mktable.py` on each table CSV, written to `lib/reports/src/figures/`

4 What this changes

HE-ML keeps its record on encryption in reports from now on. A new result goes into a report of its own under its slug, and the search log of the study is the searches section of `lib/literature/positioning.md`. The report narrows reduction to CKKS to semi-honest coalitions until prob-threshold-ckks-realizes-releases is closed, and leaves every other claim as it stood.

Table 1. The open problems HE-ML carries. The full entry stands under the same slug in the migrated problem register.

Entry	Statement	Status
prob-threshold-ckks-realizes-releases	Does threshold CKKS with Gaussian smudging realize the ideal releases of thm-dataflow-composition, under RLWE against a coalition that follows the protocol, and with an abort set against a deviating one?	Open. It needs a simulator for smudged partial decryptions and, against deviation, proofs of correct partial decryption for CKKS.
prob-he-oft-label-queries-against-head	Do the coalition's label queries in HE-OFT reach what the shared head reveals, as a collapse or a separation at the distinguish goal? Asked by Küpçü on 2026-10-16.	Open.
prob-he-oft-count-steering-bound	Can counts and norm bounds proven under encryption, as in ACORN, bound the steering of prop-he-oft-count-steering?	Open, literature unchecked.
prob-he-oft-smudging-simulation-security	Does the per-party smudging of the first theorem of HE-OFT meet standard simulation security, given fact-hhkls26-per-party-smudging-fails?	Open, literature unchecked.
prob-hlss25-smudging-simulation-security	Does the proof of HLSS25 against deviating decryptors, which smudges per party, meet standard simulation security after HHKLS26?	Open, literature unchecked.

Table 2. The results HE-ML carries, with the status each has in the migrated result register.

Entry	Statement	Status
thm-dataflow-composition	A client isolated behind Gaussian releases has the divergence of the views bounded by a Gaussian term plus an exceedance probability that coupled runs certify. The ciphertexts of an aggregation reduce to the releases up to the realization error.	Proven given the realization error. Heuristic, that threshold CKKS realizes the releases.
def-distinguish-goal	Two hypotheses on the clients' datasets. The advantage is the gap in the probability of outputting one, against the best advantage from public knowledge alone.	A definition.
thm-he-oft-semi-honest-simulation	A semi-honest server with t minus 1 clients is simulated from the ideal leakage under IND-CPA and a smudging hypothesis, which the measured configuration does not meet.	Proven in KSK26, unchecked here.
thm-he-oft-deviating-clients-bound	Against t minus 1 deviating clients and an honest server, the distinguishing advantage is a negligible term plus what the label queries give.	Proven in KSK26 under three readings of deviating messages that no proof enforces, unchecked here. No steering claim.
thm-he-oft-label-view-ordering	The label answers of t minus 1 deviating clients, at most Q per client, give no more than both shared heads in plaintext.	The ordering is proven in KSK26, unchecked here. The near-zero white-box advantage is empirical.
thm-he-oft-no-deviating-server	A client's decision to contribute a decryption share cannot depend on the plaintext, so no protocol in which clients decrypt what the server presents admits a deviating server.	Proven in KSK26, unchecked here.
prop-he-oft-count-steering	Nothing ties a client's uploaded per-class counts to its data, so a large reported count moves that row of the shared head as close to the client's head displacement as it likes.	Proven from the aggregation formula of HE-OFT, unchecked.

Table 3. The facts about prior work HE-ML carries, with how each was read.

Entry	Statement	Status
fact-he-oft-semi-honest-simulation	The record of thm-he-oft-semi-honest-simulation in KSK26.	Verified, KSK26 read in full by 2026-10-16.
fact-he-oft-deviating-clients-label-view	The record of thm-he-oft-deviating-clients-bound and thm-he-oft-label-view-ordering in KSK26.	Verified, full read.
fact-he-oft-no-deviating-server	The record of thm-he-oft-no-deviating-server in KSK26.	Verified, full read.
fact-hhkls26-per-party-smudging-fails	HHKLS26 (ASIACRYPT 2026) prove that independent per-party smudging fails standard simulation security.	Recorded from a sweep coder's extraction. Not read.
fact-hlss25-per-party-smudging-proof	HLSS25 (CCS 2025) give a proof against deviating decryptors that smudges per party.	Recorded from a sweep coder's extraction. Not read.
fact-threshold-fhe-active-security-no-ckks	Active security of threshold FHE decryption was found for thresholdized BFV, BGV and FHEW, and for no variant of CKKS.	Recorded from one search summary. No paper opened.